

BASIC ELECTRONICS

(24B11EC111)

LECTURE - 3

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Common Emitter Configuration

- Fig. 1 shows npn transistor biased in Common Emitter configuration.
- Emitter is common to both input and output.
- Input current is I_B , input voltage is V_{BE} , Output current is I_C and output voltage is V_{CE} .

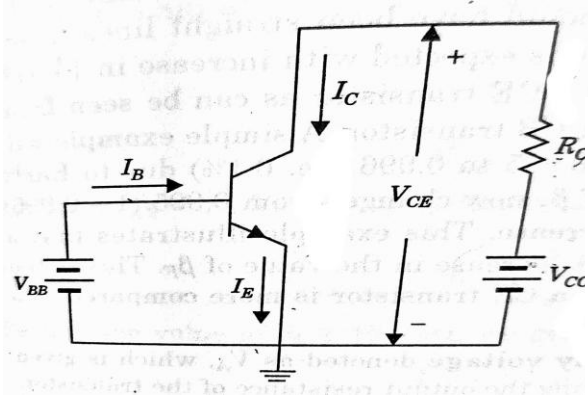


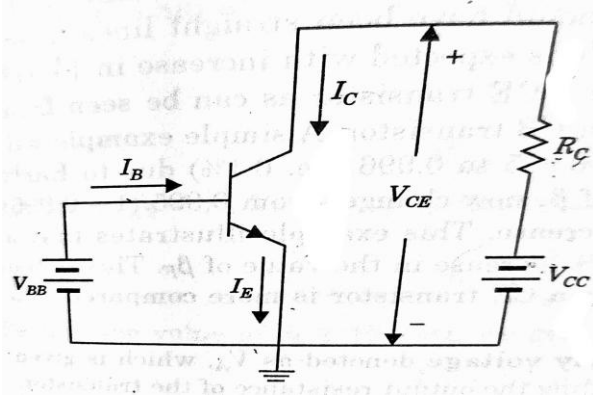
Fig. 1 Common Emitter Configuration[1]

Biasing:

Base-Emitter Circuit: A small voltage V_{BB} is applied between the base and emitter to forward bias the base-emitter junction. This allows current to flow from the base to the emitter, creating the base current I_B .

Collector-Emitter Circuit: The collector-emitter junction is reverse biased by applying a voltage V_{CC} between the collector and emitter. This allows a small base current to control a much larger collector current I_C , which flows from the collector to the emitter through the load resistor R_C .

Cont.



emitter junction is forward biased, it allows electrons (in the case of an NPN transistor) to flow from the emitter into the base region.

Since the base region is very thin and lightly doped, most of these electrons (approximately 95-99%) are swept into the collector due to the reverse bias applied at the collector-emitter junction.

This movement of electrons constitutes the **collector current** I_C , which is much larger than the base current I_B .

The remaining small portion of electrons recombines with holes in the base, creating the base current I_B .

Output Characteristics

- Fig.2 shows the output characteristics of CE configuration.
- Drawn between output current(I_C) and output voltage(V_{CE}) for various fixed values of I_B .
- The region to the right of vertical line drawn at $V_{CE} = V_{CE,sat}(\sim 0.2 \text{ V})$ and above $I_B = 0$ defines the active region.
- The region to the left of vertical line drawn at $V_{CE} = V_{CE,sat}(\sim 0.2 \text{ V})$ defines saturation region

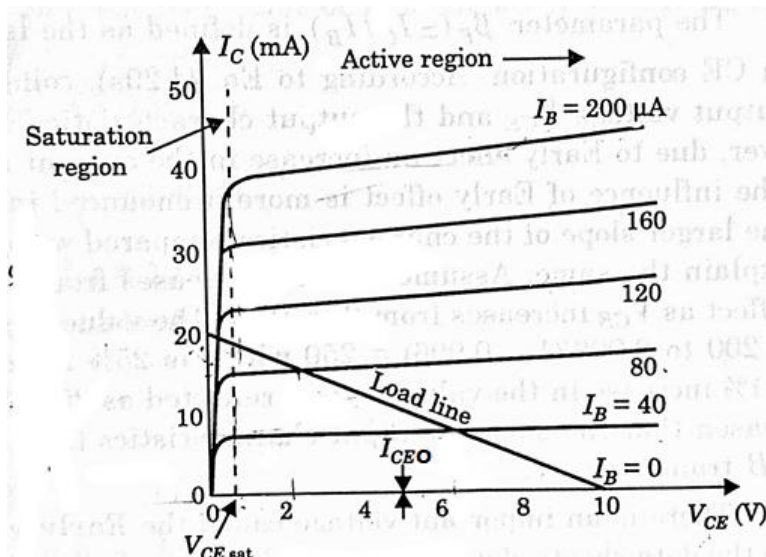


Fig. 2 Output characteristics of CE Configuration[1]

Active region

- ▶ $I_C = \alpha_F I_E + I_{CO}$ Eq. 1
- ▶ $I_E = I_C + I_B$ Eq.2
- ▶ Substituting Eq. 2 in Eq. 1
- ▶ $I_C = \frac{\alpha_F}{1-\alpha_F} I_B + \frac{I_{CO}}{1-\alpha_F}$Eq. 3
- ▶ Parameter β_F is introduced where β_F
- ▶ $\beta_F = \frac{\alpha_F}{1-\alpha_F}$ is defined as large signal current gain of transistor in CE
- ▶ configuration.
- ▶ Eq. 3 can be re written as
- ▶ $I_C = \beta_F I_B + (1+\beta_F) I_{CO}$ Eq. 4
- ▶ As $I_{CO} \ll I_B$, $I_C \approx \beta_F I_B$

Cont.

- According to eq. 4, I_C is independent of V_{CE} .
- But, in output characteristics I_C rises with a finite slope as V_{CE} increases.
- This is due to Early Effect which is more pronounced in CE configuration.
- For a 1% increase in α_F there is 25% rise in β_F . So, the slope of output characteristics is larger in case of CE configuration when compared to CB configuration.

Early Voltage (Base Width Modulation)

- ✦ In a **common-emitter configuration**, the collector current (I_C) ideally should be independent of the **collector-emitter voltage** (V_{CE}).
- ✦ However, in real-world BJTs, as V_{CE} increases, the depletion region at the collector-base junction widens, which reduces the effective base width. This is known as **base-width modulation**.
- ✦ As the base width decreases, the current increases because the recombination of charge carriers in the base is reduced.
- ✦ This results in a slight dependence of the collector current on V_{CE} , which is called the **Early effect**.

Early voltage

- It is denoted by V_A and given in datasheets of the transistor.
- Helps to find out the output resistance of the transistor.
- It is determined from output characteristics drawn for various values of V_{BE} as shown in fig. 3.
- If the linear portion of output curves are extended back, they all meet at same point on x-axis, this point gives the value of early voltage.
- Output resistance $r_o = V_A/I_C$.

The **Early voltage** V_A is a key parameter in determining the extent of the Early effect. A larger V_A means that the slope of the output characteristics is flatter (i.e., less Early effect), resulting in a more ideal transistor behavior with higher output resistance and better gain stability.

In modern transistor design, increasing V_A (by optimizing the transistor's physical structure) can mitigate the Early effect and improve performance.

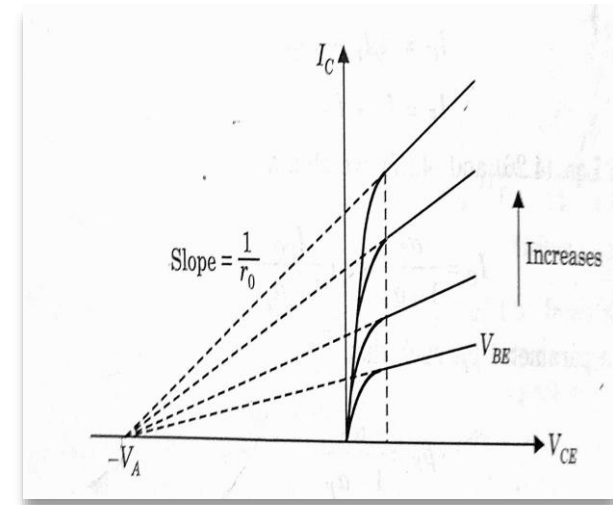


Fig. 3 Finding out early voltage from Output characteristics[1]

Cut – off mode

- If $I_B = 0$ in eq. 4,
$$I_C = (1 + \beta_F) I_{CO} = I_{CEO}$$
- I_{CEO} is the current flowing from collector to emitter with base kept open.
- As α_F is close to unity, β_F is very large and I_{CEO} is not too low to be neglected. So, transistor can't be said in cut-off region.
- For Si transistor, EB junction is short circuited and for Ge transistor EB junction is supplied with a small reverse bias of -0.1 V, so that $I_E = 0$ and $I_C = I_{CO}$ as shown in fig. 4(a) and 4(b).
- This drives the transistor in cut-off region

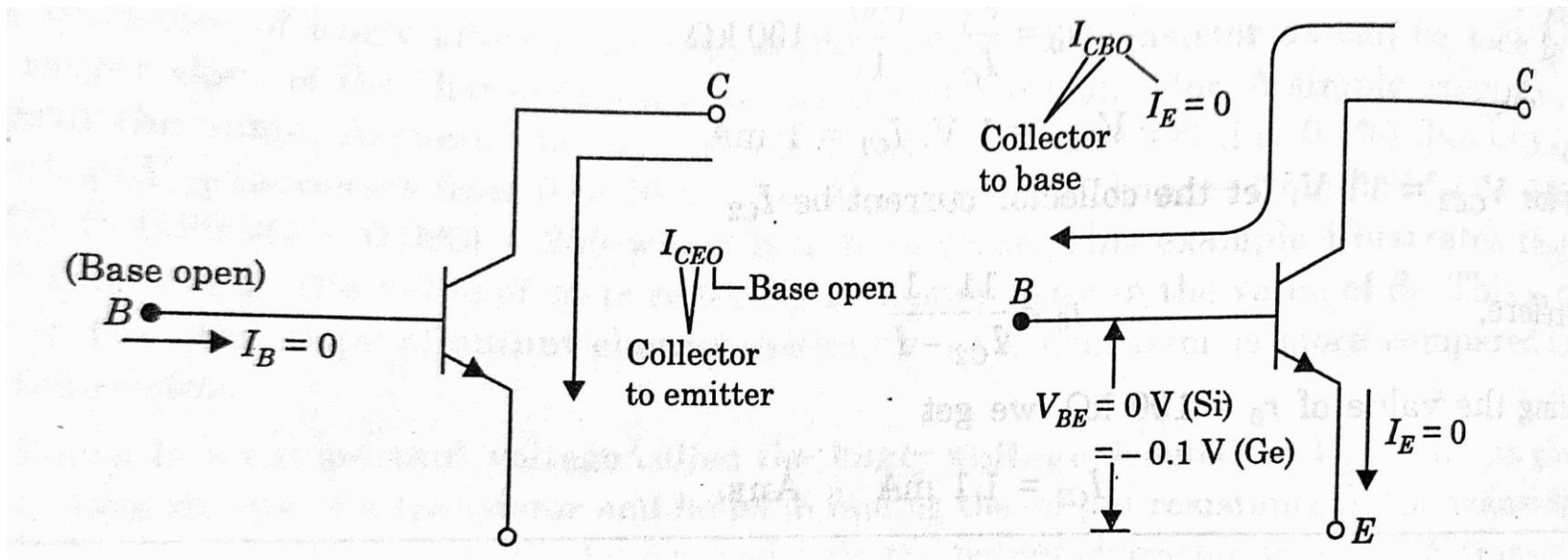


Fig. 4(a) Collector current with base open[1]

Fig. 4(b) Arrangement to drive transistor in cut-off[1]

Saturation mode

- Both EB and CB junctions are forward biased.
- Under active mode, typical cut-in voltages of EB and CB junctions are 0.7 V for Si transistor(0.2 V for Ge Transistor)
- Barrier potential of EB junction in saturation mode is a little higher than in active mode (0.8 V and 0.3 V approx. for Si and Ge transistor respectively)
- CB junction requires approximately 0.6V to forward bias. The difference in cut-in voltages of EB and CB junction are due to difference in doping profiles.
- $V_{CE, sat} = V_{CB} + V_{BE} \Rightarrow -0.6 + 0.8 = 0.2 \text{ V}$

- Applying KVL in output circuit,

$$I_C = (V_{CC} - V_{CE}) / R_C$$

- If the transistor is biased with fixed V_{CC} and R_C . A line is drawn on the output characteristics with slope = $(-1/R_C)$ as shown in fig. 2
- This is called load line and the points where it intersects with output characteristics curve, gives the operating values of I_C and V_{CE} at various I_B values.
- If the region to the left of $V_{CE,Sat}$ is magnified, this load line can be approximated as a flat line as shown in fig. 5
- For $V_{CE} < V_{CE,Sat}$, the operating value of collector current is approximately same, irrespective of the value of base current. Thus, collector current is said to be saturated.

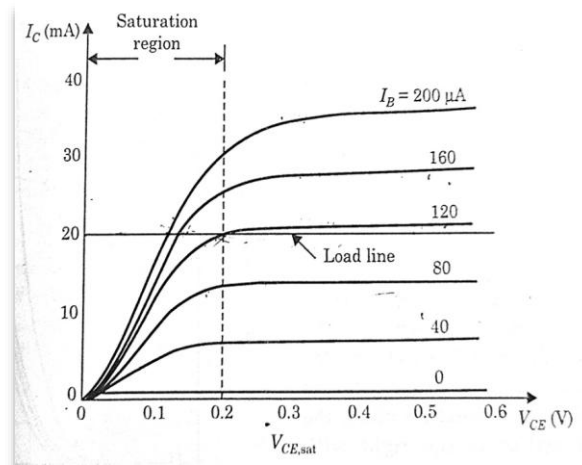


Fig. 5 Saturation region magnified[1]

Now,

$$I_{C,Sat} = (V_{CC} - V_{CE,Sat})/R_C$$

Minimum base current required to drive transistor in saturation is:

$$I_{B,min} = I_{C,Sat}/\beta_F$$

- To operate transistor in saturation mode, the circuit should be designed such that I_B is greater than $I_{B,min}$ by a factor of 2 to 10.
- If I_B is increased above $I_{B,min}$, β will reduce as $I_{C,sat}$ is constant. The new value of β is known as β_{forced} where

$$\beta_{forced} = I_{C,Sat}/I_B$$

Input Characteristics

- Fig. shows the input characteristics of CE configuration.
- It is a family of curves drawn between input voltage (V_{BE}) and input current (I_B) for different values of output voltage (V_{CE})
- For $V_{CE} = 0$, the characteristics represent forward biased EB junction.
- As V_{CE} increases (keeping constant V_{BE}), V_{CB} increases and hence the CB junction reverse bias also increases.
- Due to Early effect, I_B decreases as effective base width reduces.
- This results in shifting of input characteristics towards right with increase in V_{CE} .

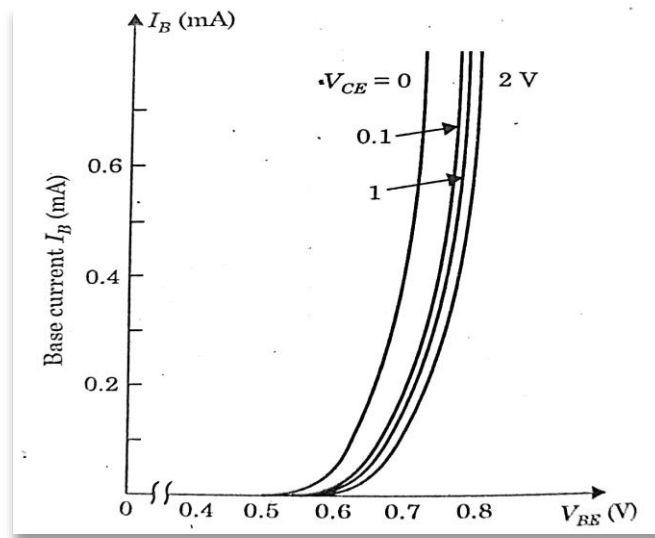


Fig. 5 Input characteristics of CE configuration[1]

References

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- R. Boylestad and L. Nashelsky, ‘Electronic Devices and Circuit Theory’, PHI, 7e, 2001.